

Akash Kothari

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Computer Engineering (Co-op)

Waterloo, ON

GPA: 4.0/4.0

EXPERIENCE

AI/ML Powered Software Engineering Intern

Jan. 2026 - April. 2026

RamSoft

Toronto, ON

- Reduced transcription **word error rate by 80%** across 200 clinical audio samples by building a **MedASR** pipeline with chunked streaming, PyTorch word-confidence scoring, and a chunk-replay simulator for iterative evaluation, containerized to **ACR** and deployed via **Kubernetes**.
- Designed a **Cosmos DB** → **SQL Server** schema migration for a scheduling epic, producing normalized **Appointment** and **Reservation** table definitions with wall-clock time storage and full recurrence support.
- Built an internal **MCP server** for **Azure SQL** with MFA token auth, enabling natural-language database queries

Cloud Software Engineering Intern

Jun. 2025 - Sep. 2025

Tripleview Technologies

Mississauga, ON

- Automated Teams call recording retrieval using **Azure**, **Microsoft Graph**, and **C#**, eliminating **10+ hours/week** of manual work for customer support teams.
- Built an **AI summarization pipeline** with webhook triggers and polling logic, processing **8+ transcripts/hour** into structured **Zendesk** tickets and reducing team miscommunication.
- Owned the full dev lifecycle delivering **production Azure tooling** now expanded across internal teams.

Software Developer (Contract)

Jan. 2026 - Jun. 2026

Mercury Mortgages

Mississauga, ON

- Built a full-stack expense system processing **600+ receipts** and **450+ transactions** across **19 sub-companies**, automating receipt-to-transaction matching and exportable reports, **eliminating a manual bookkeeping role**.
- Engineered a **dual-ingestion pipeline** parsing **250+ emails** via **Microsoft Graph** webhooks, with async processing via a **Redis** job queue and threaded workers for PDF receipt conversion.
- Implemented a **rules engine** mapping vendor and location data to GL codes, with **role-based access control** and admin-gated expense report approvals.

PROJECTS

QuikMart - Grocery Delivery Startup | *NestJS, TypeORM, PostgreSQL, Next.js, React* May 2026 - Present

- Built a **bulk CSV product upload system** processing up to **1,000 products** per upload with server-side validation, flexible column aliasing, category name matching, and a two-step validate-then-import flow.
- Implemented **tenant isolation** with ownership-based authorization, and designed a product **recommendation scoring engine** with weighted signals (popularity, repeat purchases, location) and cold-start fallbacks.
- Wrote **80+ tests** including integration tests with concurrent idempotency and email outbox processor lifecycle against a real **PostgreSQL** test database.

FocusOS - Personal Productivity OS | *FastAPI, PostgreSQL, React, Azure OpenAI, Docker*

- Built a full-stack productivity OS with an **AI job analysis pipeline** using **Azure OpenAI** — scores resume-job match, extracts ATS keywords, and generates tailored LaTeX resumes.
- Designed a **multi-step background task pipeline** with non-blocking AI calls, retry logic, and real-time status polling, deployed with **Docker** and a **PostgreSQL** backend.

Mathora - Learn Math Visually | *React Three Fiber, Three.js, LLMs, Express*

- Built an LLM-controlled math engine driving **15+ animation types** across 3D graphing, whiteboard, camera, and voiceover via structured prompt schemas, with **10 precomputed templates** for instant rendering.
- Reduced perceived latency from **30s to under 2s** by streaming animation steps via **SSE** as they were generated

TECHNICAL SKILLS

Languages: Python, C/C++, C#, Java, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Next.js, NestJS, FastAPI, Flask, Node.js/Express, Three.js, React Three Fiber

Cloud & DevOps: Azure, GCP, Kubernetes, Docker, PostgreSQL, CI/CD, Redis, REST APIs, Git, Jira

ML/Data: Scikit-learn, PyTorch, XGBoost, Pandas, Feature Engineering, Computer Vision, NumPy, SciPy